

Formatvie Quiz: Derivatives of Exponential & Logarithmic Functions

K/U	A	C	T
/16	/8	/6	/8

KNOWLEDGE / UNDERSTANDING

1. Find the first derivative: (fully factored with positive exponents)

a) $y = (3x - 5)e^{3x}$ ✓

b) $y = \log_6(5x)$ ✓

c) $y = \ln(2x^2 - 1)^3$ ✓✓

d) $y = \ln(\ln x^2)$ ✓✓

e) $y = 5^{2x-4}(2x-4)$ ✓✓

f) $y = e^{\sin x^2}$ ✓✓

g) $f(x) = \cos(x+1)\sin^2(x^2-3x)$ ✓✓✓

h) $y = x^3 \log_3(x^3 - 2x + 3)$ ✓✓✓

APPLICATION

2. Find the equation of the tangent to the curve $y = 3^x$ at the point where $x = -2$ ✓✓✓✓

3. Find the equation of the tangent to the curve $y = \ln(1 + e^x)$ which is parallel to the line $x - 2y = 0$. ✓✓✓✓

COMMUNICATION

4. a) State how to find the derivative of $y = a^{f(x)}$ in your own words. ✓✓
- b) State how to find the derivative of $y = e^{f(x)}$ in your own words. ✓✓
- c) Explain how the rule in b) is a special case of the rule in a). ✓✓

THINKING, INQUIRY AND PROBLEM SOLVING

5. The function $f(t) = 2000e^{3t}$ represents the number of bacteria after t hours. Find the rate of change of growth when the population of bacteria is at 800,000. ✓✓✓✓

6. If $y = x^2e^x$ prove that $y''' = e^x(x^2 + 6x + 6)$ ✓✓✓✓